ABSTRACT

The increase in population, has led to tremendous degradation in the state of affairs of hygiene with respect to waste management system. The spillover of waste in civic areas generates the polluted condition in the neighboring areas. It may aggravate numerous severe diseases for the nearby people. This will humiliate the appraisal of the affected area. For eliminating or mitigating the garbage's and maintains the cleanness, it requires smartness-based waste management system. This is proposed on IT based smart waste clean management system which checks the waste level over the dustbins by using Sensor systems. Once it detected immediately this system altered to concern authorized person. To monitor and integrate an android application is developed for the desired information which is related to the various level of waste in different locations. This is ensued the greenish in the environment and support for Swachh Bharat for cleanness.

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LIST OF ABBREVIATIONS

ADC	-	Analog to Digital Converters
API	-	Application Programming Interface
AWP	-	Area Waste Plan
AWTT	-	Alternative Waste Treatment Technologies
EDIE	-	Enivornmental Data Interactive Exchange
EIA	-	Enivornmental Impact Assessment
GUI	-	Graphical User Interface
GND	-	Ground
GSM	-	Global System for Mobile Communications
GAYT	-	Get Us You Throw
IDE	-	Integrated Development Environment
IR	-	Infrared
LCD	-	Liquid Crystal Display
LED	-	Light Emitting Diode
PWM	-	Pulse Width Modulation
RFID	-	Radio Frequency Identification
RSA	-	Recycled and Secondary Aggregate
RWM	-	Recycling and Waste Management Exhibition
RX	-	Receiver
SWA	-	Solid Waste Analysis
TX	-	Transmit
VLLW	-	Very Low Level Waste
WDA	-	Waste Disposal Authority
WMS	-	Waste Management Strategy
WRAP	-	Waste and Resources Action Programme

CHAPTER 1

INTRODUCTION

There is an implicit need of Smart waste management to improve the infrastructure of our cities. The idea of Smart Dustbin is for the Smart cities, buildings, educational institutes, Hospitals, Bus stands and any other public places. We are living in an age where tasks and systems are fusing together with the power of IoT to have a more efficient system of working and to execute jobs quickly. With all the power at our finger tips this is what we have come up with. One of the main concerns with our environment has been solid waste management which impacts the health and environment of our society. The detection, monitoring and management of wastes is one of the primary problems of the present era. The traditional way of manually monitoring the wastes in waste bins is a cumbersome process and utilizes more human effort, time and cost. It can be easily avoided by our present technologies. In this project were bring new direction for integrating IoT for green environment by cleaning the waste automatically and to provide a more efficient solution.

.

CHAPTER 2

LITERATURE SURVEY

[1] P. Arunadevi, S.Karthika, V.V.Swetha raj and K.Aamir suhail (2019), 'Smart garbage monitoring system using Internet of things', International Research Journal of Engineering and Technology (IRJET).

The literature surveyed some different papers to get information about the existing work which have been done. For the garbage detection, some paper uses weight sensor. It gives the weight of the garbage in the dustbin. But it doesn't provide any information about the level of the garbage in the dustbin. Hence, therefore used Infrared Sensor (IR) for garbage detection. IR Sensor radiates light which is invisible to the human eye because it is at infrared wavelengths, but it can be detected by electronic devices. IR transmitter consists of LED which sends the IR beam. Technology uses IR sensor, Microcontroller, Graphical user interface (GUI).The equipment is placed on the side walls of the bin which consists of two parts: one is the receiver-transmitter and sensor. Sensor is used to indicate the level of the bin which is connected to the transmitter that transmits a signal of fullness of the bin tothe receiver at the server host. A manager is appointment at the server side whose job is to find the shortest route and intimate it to the trucker driver to collect it in short interval of time.

[2] Ruhin mary saji, Drishya gopakumar ,Mohammed sayed, Lakshmi and Harishkumar (2016), 'A Survey of smart garbage management in cities using IoT',International Journal of Engineering and Computer Science

The garbage management in cities has to be effectively and efficiently implemented. The various proposals were put forward and some of them already implemented. But it cannot be considered as an effective one. So a survey was done among different proposals and this survey paper includes survey among different methods for smart garbage management in cities using IoT.

The paper Smart Garbage Management in Smart Cities using IoT proposed a method as follows. The level of garbage in the dustbins is detected with the help of ultrasonic sensors system, and communicated to the authorized control room through GSM system. Arduino microcontroller is used to interface the sensorsystem with GSM system. A GUI is also developed to monitor the desired information related to the garbage for different selected locations. This will help tomanage the garbage collection efficiently. Level detector consists of IR sensors which is used to detect the level of the garbage in the dustbin. The output of level detector is given to microcontroller. Four IR sensors are used to indicate the different levels of the amount of the garbage collected in the dustbin which is placed in public area.

[3] Neha shinde, Sayli Bhambre, Sharddha Thakur, and Varsha Devkule (2018), 'A Survey on Garbage Collection and Monitoring System for smart cities using IoT', International Research Journal of Modernization of Engineering and Science.

The idea of smart garbage bins and systems have been in discussion for quite a long time. The technologies used at disposal to develop this smart system have also evolved, Internet of Things (IoT). Each idea seems to be similar but is slightly different at its core and our proposed work is no exception from the same. After the IoT field, finding its hold in our lives, this is our original plan for designing a smart garbage collection system which has provision for citizen participation and analysis of data for better decision making. At hardware level, the smart system is a garbage bin with ultrasonic sensor, a micro-controller and Wi-Fi module for transmission of data.

The worldwide implementation of Internet of Things is possible with a Cloud centric vision. This work exploits the future possibilities, key technologies and application that are likely to drive IoT research. But a strong foundation to our work is provided, where the basics and applications of Arduino board is explained . It is quite interesting as it implements a GAYT (Get As You Throw) system concept as a way to encourage recycling among citizens. As we would discuss further, the citizen participation part of our system is quite influenced by their work.

[4] Shephali Rakhunde, Shreya Ghavghave, Shraddha Jagtap, Priyanka Chimegaokar, and J.Y.Hande (2019), 'A Review on smart garbage dustbin',International Journal of Science Research and Engineering Trend.

A Smart Waste Bin for Smart Waste Management proposed by, In this paper, the system consists of sensors to measure the weight of waste and the level of waste inside the bin. Bluetooth is attached for short range communication. The researchers suggests the method for garbage management which is as follows. In this paper, Arduino UNO to check the level of garbage filled in the dustbin and sends the alert to the municipal web server once if garbage is filled. After cleaning the dustbin, the driver confirms the task of emptying the garbage with the aid of RFID Tag. RFID is a computing technology that is used for verification process and in addition, it also enhances the smart garbage alert system by providing automatic identification of garbage filled in the dustbin and sends the status of clean-up to the server affirming that the work is done. The researchers suggests the method for garbage management which is as follows. In this paper the system makes use of Arduino Uno board, LCD screen, GSM modem for sending data. The system is powered by a 12V transformer. The LCD screen is used to display the status of the level of garbage collected in the bins. Whereas GSM is built to show the status to the user, monitoring it with SMS. The SMS consists of text related to all garbage bins. The LCD screen shows the status of the garbage level. The system puts on LCD screen continuously monitoring of garbage with Arduino board. The researchers suggests the method for garbage management which is as follows.

CHAPTER 3

EXISTING SYSTEM

- ✓ The existing system has the limitations as time consuming, trucks goand empty the containers, even they are empty.
- ✓ The cost is high with unhygienic environment. Even the bad odour causes the unhealthy environment.
- ✓ In Existing Implementation of real time waste management system using GSM system.
- ✓ These systems use automation technique reduces human efforts.
- ✓ Ultrasonic sensor is provided for detection of garbage levels and passes signal to the microcontroller which is finally given to the GSM system.
- ✓ GSM module sends message to the authorized person.

CHAPTER 4

PROPOSED SYSTEM

- ✓ The implementation starts by setup ESP8266 by flashing the latest version of the firmware.
- ✓ In this project we can sense the garbage level and reduce the disesases,human effort,time.
- ✓ The ultrasonic sensor transmit/receive the signal from dustbin and the data send to the NodeMCU and store in cloud.
- ✓ Then the ESP8266 is connect with thingspeak app.The app shows the graph representation of the garbage.
- ✓ System can be accessed anytime and from anywhere.
- ✓ Real-time data transmission and access.
- ✓ Avoids the overflows of garbage bins.

4.1 Advantages

- Ultrasonic sensor accuracy is high.
- To reduce the human effort,cost,time.
- To reduce the diseases.
- Real time information on the fill level of the dustbin.
- Improves Environment quality.

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

This proposed work of “Smart garbage monitoring system using IoT” will use both hardware and software modules. To perform the monitoring a garbage and to analyze the garbage waste level and it uses the Nodemcu,ESP8266 and sensors.

5.1 HARDWARE MODULES

- NodeMCU(ESP8266)
- Ultrasonic sensor
- 9V HW battery
- Jumper wires
- Garbage bin

5.1.1 NODEMCU(ESP8266) SPECIALIZATION

✓ Microcontroller	ESP-8266 32 bit
✓ NodeMCU Model	Amica
✓ NodeMCU Size	49mm x 26mm
✓ Carrier Board Size	n/a
✓ Pin Spacing	0.9" (22.86mm)

✓ Clock Speed	80 MHz
✓ USB to Serial	CP2102
✓ USB Connector	Micro USB
✓ Operating Voltage	3.3V
✓ Input Voltage	4.5V-10V
✓ Flash Memory/SRAM	4 MB / 64 KB
✓ Digital I/O Pins	11
✓ ADC Range	0-3.3V
✓ UART/SPI/I2C	1 / 1 / 1
✓ WiFi Built-In	802.11 b/g/n
✓ Temperature Range	-40C - 125C

5.1.2 NodeMCU(ESP8266)

NodeMCU is an open source platform based on ESP8266 which can connect objects and let data transfer using the Wi-Fi protocol. In addition, by providing some of the most important features of microcontrollers such as GPIO, PWM, ADC, and etc, it can solve many of the project's needs alone.

The general features of this board are as follows:

- Easy to use
- Programmability with Arduino IDE or IUA languages.
- Available as an access point or station.
- practicable in Event-driven API applications.
- Having an internal antenna.



Figure 5.1 NodeMCU(ESP8266)

5.1.3 Ultrasonic sensor

An ultrasonic sensor is an electronic device that measures the distance of a target object by emitting ultrasonic sound waves, and converts the reflected sound into an

electrical signal. Ultrasonic waves travel faster than the speed of audible sound (i.e. the sound that humans can hear). Ultrasonic sensors have two main components: the transmitter (which emits the sound using piezoelectric crystals) and the receiver (which encounters the sound after it has travelled to and from the target).

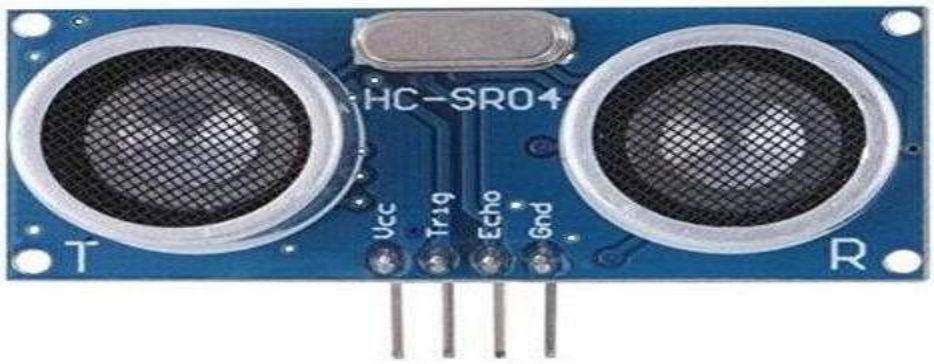


Figure 5.2 Ultrasonic sensor

5.1.4 9V HW Battery

Hi-Watt 9V Battery is the most commonly used and portable 9V battery. It is non-rechargeable and is a high capacity and low-cost solution for many electronic devices. It is based on Zinc Carbon Chemistry and can be used easily replaced if discharged just like any standard AA and AAA batteries. The battery can be used to power LEDs, Toys, Flashlight and Torch, electronic equipment like multimeter, wall clocks, or other devices with a 9V system.



Figure 5.3 9V HW Battery

5.1.5 Jumper wires

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. Fairly simple. In fact, it doesn't get much more basic than jumper wires. Jumper wires typically come in three versions:

- male-to-male,
- male-to-female
- female-to-female.



Figure 5.4 Jumper wires

5.1.6 Garbage bin

Maintaining a clean and tidy environment is vital to the comfort of your customers, guests, and employees. You must choose the appropriate trash cans and garbage bins for your affirmation to maintain cleanliness.

The Importance of Garbage Bin are:

- To Maintain a safe environment.
- Recycling.
- A clean environment.



Figure 5.5 Garbage bin

5.2 SOFTWARE REQUIREMENTS

- ✓ Operating system : Windows, Linux
- ✓ IDE : ARDUINO

5.2.1 SOFTWARE DESCRIPTION

- ✓ **IDE : ARDUINO**

The Arduino integrated development environment (IDE) is a cross- platform application (for Microsoft Windows, macOS, and Linux) that is written in the Java programming language. It originated from the IDE for the languages Processing and Wiring. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax

highlighting, and provides simple one-click mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus. The source code for the IDE is released under the GNU General Public License, version 2.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub main() into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

From version 1.8.12, Arduino IDE windows compiler supports only Windows 7 or newer OS. On Windows Vista or older one gets "Unrecognized Win32 application" error when trying to verify/upload program. To run IDE on older machines, users can either use version 1.8.11, or copy "arduino- builder" executable from version 11 to their current install folder as it's independent from IDE.object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM BLOCK DIAGRAM

In this system, the ultrasonic sensor sensing garbage level and transfer a data to NodemcuESP8266 and then ESP8266 connect with thingspeak. The thingspeak shows the graph representation of our garbage level .Figure 5.1 shows the overallblock diagram of the system.

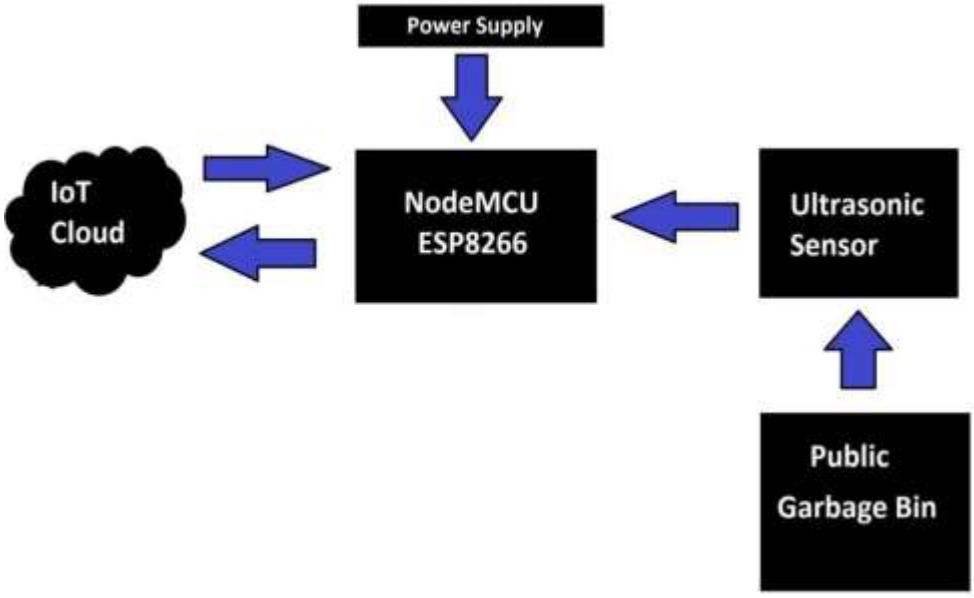


Figure 6.1 Block Diagram of the Proposed System

CHAPTER 7

TESTING

7.1 TESTING DESCRIPTION

Testing is a series of different tests that whose primary purpose is to fullyexercise the computer-based system. Although each test has a different purpose, all work should verify that all system element has been properly integrated and performed allocated function. Testing is the process of checking whether the developed system works according to the actual requirement and objectives of thesystem. The philosophy behind testing is to find the errors. A good test is one thathas a high probability of finding an undiscovered error. A successful test is one that uncovers the undiscovered error. Test cases are devised with this purpose in mind. A test case is a set of data that the system will process as an input.

System Testing

After a system has been verified, it needs to be thoroughly tested to ensure that every component of the system is performing in accordance with the specific requirements and that it is operating as it should include when the wrong functionsare requested or the wrong data is introduced. Testing measures consist of developing a set of test criteria either for the entire system or for specific hardware, software and communications components. For an important and sensitive system such as an electronic voting system, a structured system testing program may be established to ensure that all aspects of the system are thoroughly tested.

Test measures for hardware may include:

- ✓ Applying “non-operating” tests to ensure that equipment can

stand upto expected levels of physical handling.

- ✓ Testing “hard wired” code in hardware (firmware) to ensure its logicalcorrectness and that appropriate standards are followed.

Tests for software components also include:

- ✓ Testing all programs to ensure its logical correctness and that appropriate design, development and implementation standards have been followed.
- ✓ Conducting “load tests”, simulating as close as possible a variety of “real life” conditions using or exceeding the amounts of data that couldbe expected in an actual situation.
- ✓ Verifying that integrity of data is maintained throughout its required manipulation.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 OUTPUTS

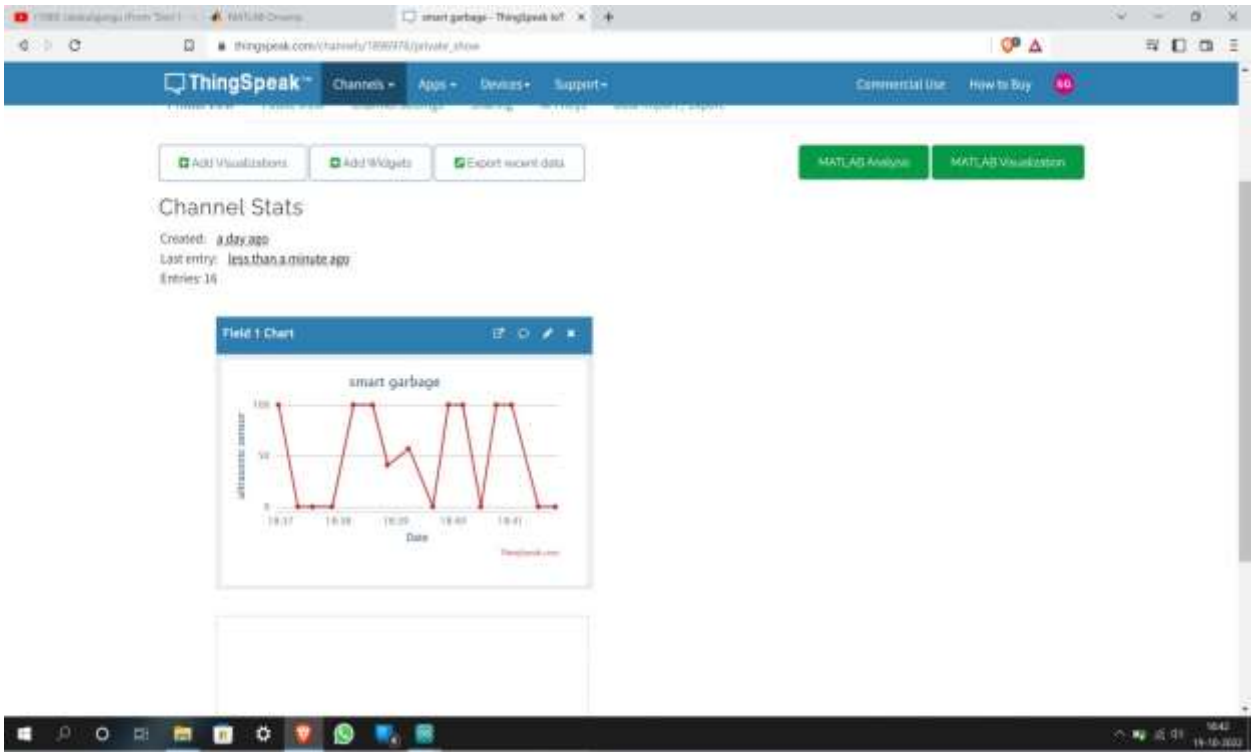


Figure 8.1 Garbage level graph representation.

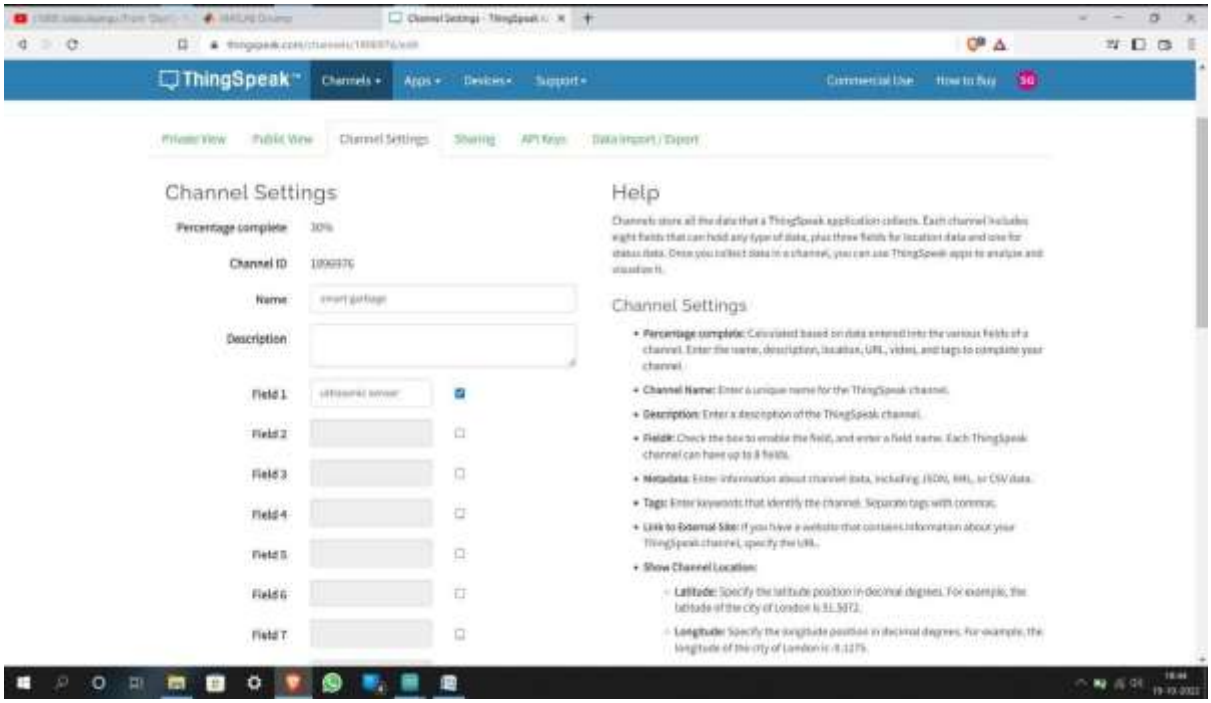


Figure 8.2 Channel and field setup in Thingspeak.

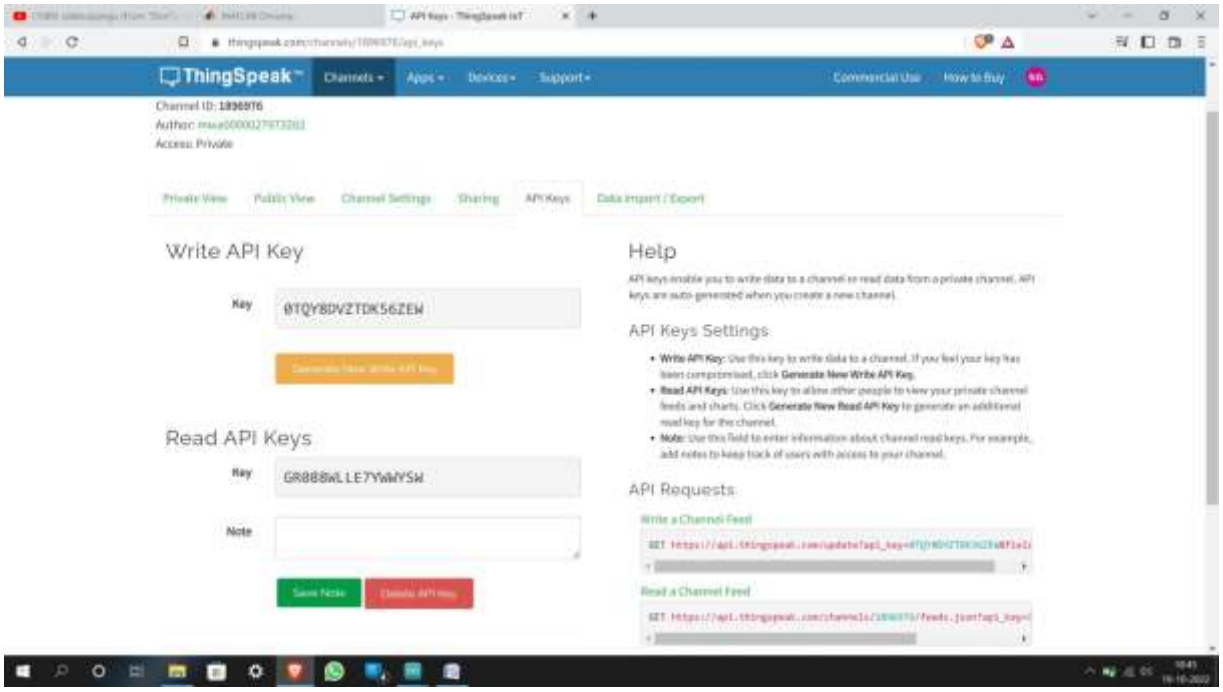


Figure 8.3 Thingspeak channel write and read API keys.

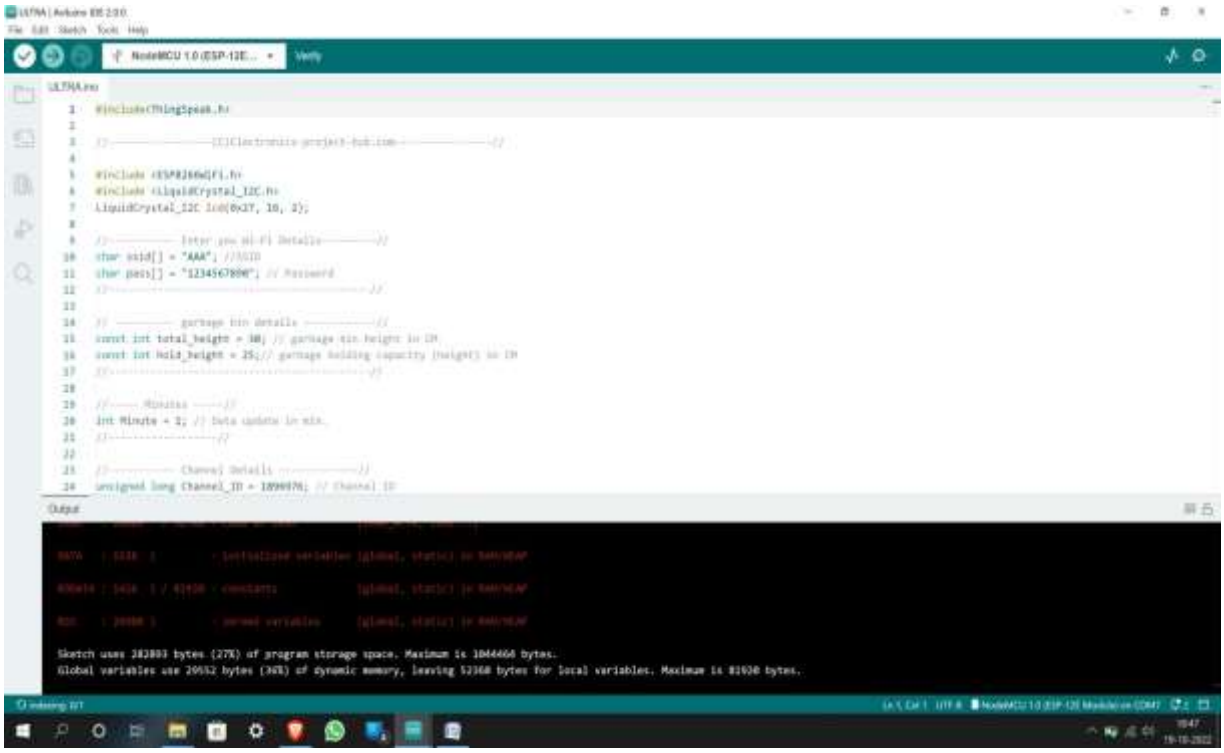


Figure 8.4 NodeMCU program compiling from Arduino IDE.

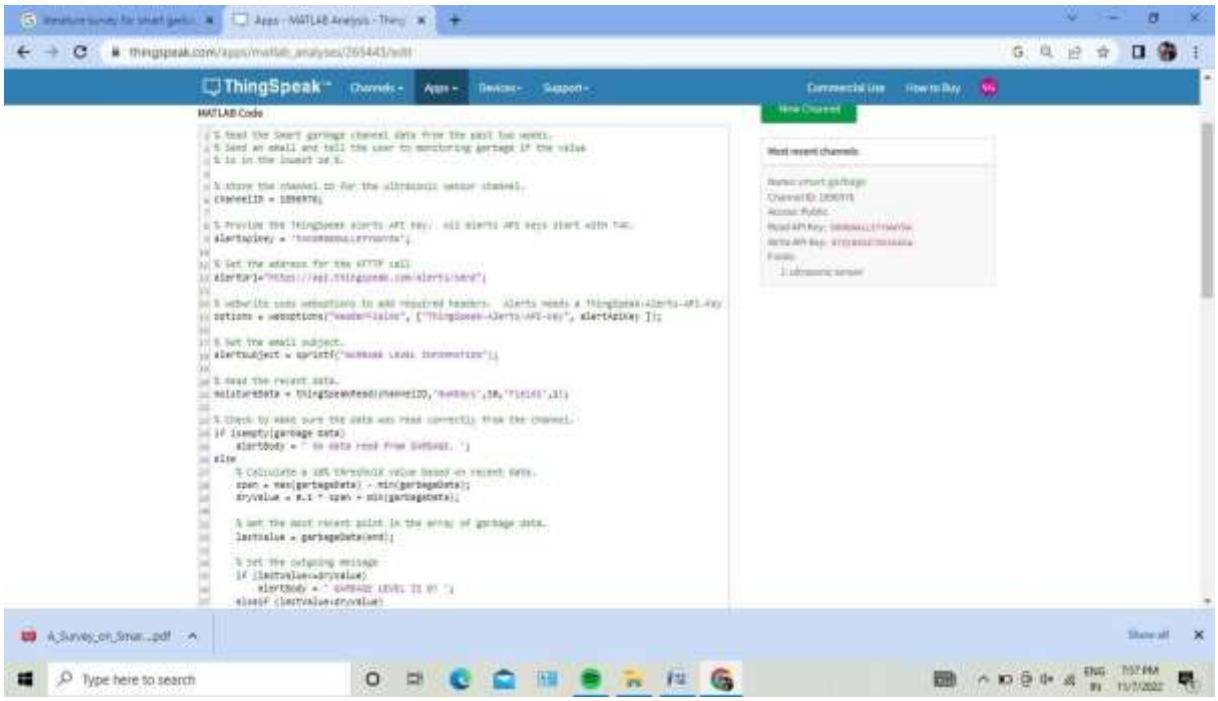


Figure 8.5 Trigger mail notification in Thingspeak.

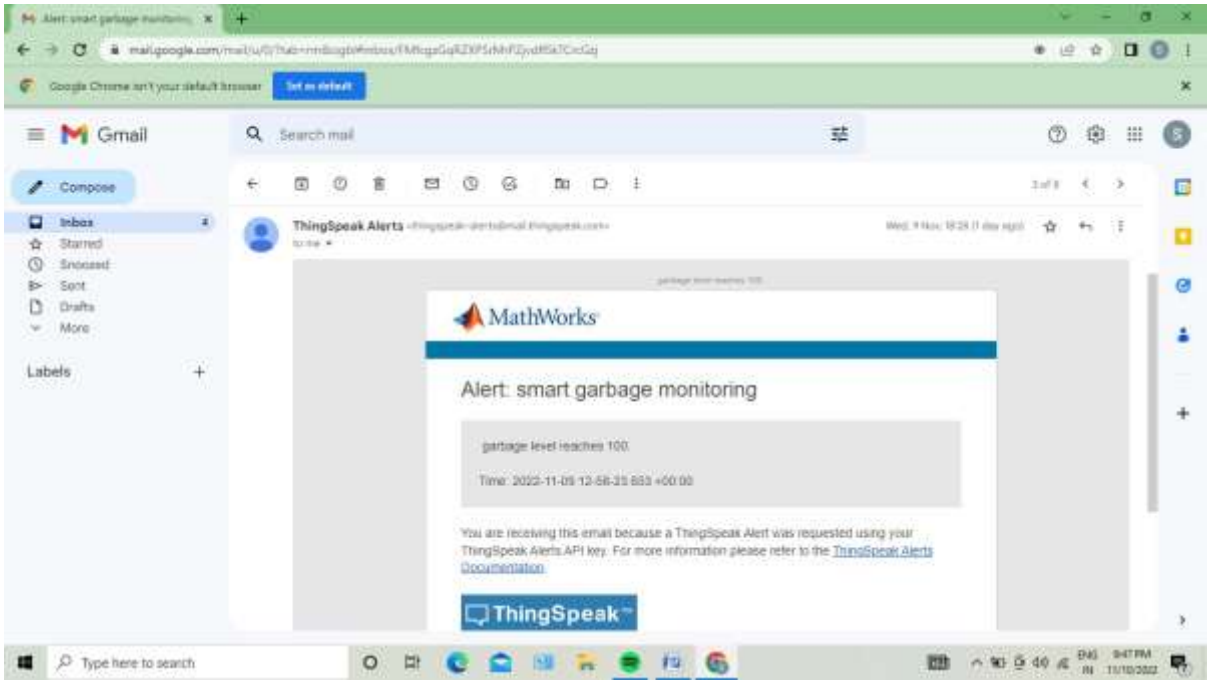


Figure 8.6 Email notification in mail.

CHAPTER 9

CONCLUSION

The main objective is to maintain the level of cleanliness in the city and form an environment which is better for living. By using this system we can constantly check the level of the garbage in the dustbins. If a dustbin has reached the maximum level then the employees can be informed and they can immediately take certain actions to empty it as soon as possible. The employees can check the status of these bins anytime on their mobile phones. This can prove to be a very useful system if used properly.

The system can be used as a benchmark by the people who are willing to take one step further for increasing the cleanliness in their respected areas. Ultrasonic sensor is being used in this system to check the level of garbage in the dustbins but in future various other types of sensors can be used with the ultrasonic sensor to get more precise output and to take this system to another level. Now this system can be used in certain areas but as soon as it proves its credibility it can be used in all the big areas. As this system also reduces manual work certain changes can be done in the system to take it to another level and make it more useful for the employees and people who are using it. In future, a team can be made which will be in charge for handling and maintaining this system and also to take care of its maintenances.

APPENDICES

```
#include <ESP8266WiFi.h>
#include <ThingSpeak.h>

//----------- Wi-Fi Configuration (Placeholders) ---------//
char ssid[] = "YOUR_WIFI_SSID";
char pass[] = "YOUR_WIFI_PASSWORD";
//------------------------------------------------------//

//----------- ThingSpeak Channel Configuration ----------//
unsigned long Channel_ID = 0000000;              // Enter your Channel ID
const char * WriteAPIKey = "YOUR_WRITE_API_KEY"; // Enter your Write API Key
const int Field_number = 1;
//------------------------------------------------------//

// ---------- Garbage Bin Physical Dimensions ----------//
const float total_height = 21.5; // Total bin height in cm
const float hold_height = 16.0;  // Holding capacity in cm
float snsr_to_max = 0.0;
//------------------------------------------------------//

// Hardware Pin Configuration
const int trigger = D5;
const int echo = D6;

long duration;
int distanceCM;
int resultCM;
int bin_lvl = 0;

WiFiClient client;
```

```
unsigned long lastUploadTime = 0;
const unsigned long uploadInterval = 60000; // Upload interval: 1 minute (60,000 ms)

void setup()
{
  Serial.begin(115200);
  pinMode(trigger, OUTPUT);
  pinMode(echo, INPUT);

  WiFi.mode(WIFI_STA);
  ThingSpeak.begin(client);

  snsr_to_max = total_height - hold_height;
  delay(1500);
}

void loop()
{
  ensureInternetConnection();
  measureDistance();

  // Non-blocking upload every 60 seconds
  if (millis() - lastUploadTime >= uploadInterval)
  {
    Serial.print("Garbage Level: ");
    Serial.print(bin_lvl);
    Serial.println("%");

    uploadData();
    lastUploadTime = millis();
  }
}
```

```
void measureDistance()
{
  digitalWrite(trigger, LOW);
  delayMicroseconds(2);
  digitalWrite(trigger, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigger, LOW);

  duration = pulseIn(echo, HIGH);
  distanceCM = duration * 0.034 / 2;
  bin_lvl = map(distanceCM, snsr_to_max, total_height, 100, 0);
  if (bin_lvl > 100) bin_lvl = 100;
  if (bin_lvl < 0) bin_lvl = 0;
}
void uploadData()
{
  ensureInternetConnection();
  int status = ThingSpeak.writeField(Channel_ID, Field_number, bin_lvl, WriteAPIKey);
  if (status == 200) {
    Serial.println("Data updated successfully.");
  } else {
    Serial.print("Upload failed, HTTP error code: ");
    Serial.println(status);
  }
}
void ensureInternetConnection()
{
  if (WiFi.status() != WL_CONNECTED)
  {
    Serial.print("Connecting to Wi-Fi...");
    WiFi.begin(ssid, pass);
    while (WiFi.status() != WL_CONNECTED)
    {
```

```
    delay(500);
    Serial.print(".");
   }
   Serial.println("\nWi-Fi Connected.");
  }
}
```

TRIGGER EMAIL NOTIFICATION CODE

```
% ThingSpeak Channel Configuration
channelID = 0000000; % Replace with your ThingSpeak Channel ID
% ThingSpeak Alerts API Key (Keep this secret)
alertApiKey = 'YOUR_THINGSPEAK_ALERT_API_KEY';
alertUrl = "https://api.thingspeak.com/alerts/send";
options = weboptions("HeaderFields", ["ThingSpeak-Alerts-API-Key", alertApiKey]);
alertSubject = "Smart Waste Management Alert: Bin Status";
% Read the recent 30 days of data from field 1
garbageData = thingSpeakRead(channelID, 'NumDays', 30, 'Fields', 1);
if isempty(garbageData)
   alertBody = 'No telemetry data received from the smart waste bin.';
else
   span = max(garbageData) - min(garbageData);
   thresholdValue = 0.1 * span + min(garbageData);
   lastValue = garbageData(end);
   if (lastValue <= thresholdValue)
      alertBody = 'Notice: Garbage bin level has reached 0% (Empty).';
   else
      alertBody = 'Alert: Garbage bin capacity threshold reached (>90% full). Dispatch
collection team.';
   end
end

% Send Email Notification
try
   webwrite(alertUrl, "body", alertBody, "subject", alertSubject, options);
   fprintf("Alert sent successfully.\n");
catch someException
   fprintf("Failed to send alert: %s\n", someException.message);
end
```

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